

COTOBATO, Philippines

WMO: 987460

Lat: 7.162N

Lon: 124.215E

Elev: 62

StdP: 100.58

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	23.9	20.4	15.1	26.8	20.9	15.7	25.9	5.4	31.9	5.1	31.6	1.4	180	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.1	34.4	26.9	33.8	26.9	33.2	26.9	27.9	33.2	27.6	32.8	27.4	32.5	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	32.4	26.0	21.5	32.3	25.7	21.2	31.9	89.6	33.5	88.3	33.0	87.2	32.7	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.8	4.3	DB	22.4	35.5	1.8	1.4	21.1	36.5	20.0	37.3	19.0	38.1	17.7	39.1	(4)
(5)			WB	20.8	28.7	0.9	0.9	20.1	29.4	19.6	29.9	19.1	30.4	18.4	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	28.2	28.6	29.0	29.2	28.8	28.3	28.0	28.1	28.0	28.3	28.4	28.5	(6)	
	DBStd	0.98	0.97	0.85	0.92	0.87	1.00	0.93	0.91	0.85	0.91	0.89	0.82	0.87	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6728	563	521	589	574	583	548	557	561	540	566	552	574	(10)	
	CDD18.3	3687	304	288	331	324	325	298	298	303	290	307	302	316	(11)	
	CDH23.3	41029	3268	3223	3850	3882	3761	3284	3161	3273	3101	3348	3361	3518	(12)	
	CDH26.7	15535	1142	1229	1588	1677	1538	1207	1083	1164	1078	1229	1265	1333	(13)	
(14) Wind	WSAvg	2.1	2.1	2.3	2.2	2.2	2.1	2.0	2.0	2.0	2.0	2.0	2.0	2.1	(14)	
(15) Precipitation	PrecAvg	2327	92	99	108	134	243	290	270	217	252	269	177	133	(15)	
	PrecMax	3510	200	263	292	398	540	621	687	363	606	614	361	425	(16)	
	PrecMin	1206	0	1	0	44	58	143	0	94	74	77	38	33	(17)	
	PrecStd	692	51	67	88	94	131	121	157	67	142	138	81	86	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	34.3	34.9	35.1	35.4	33.9	33.5	33.7	33.7	34.0	33.9	34.2	(19)	
		MCWB	26.8	26.4	26.4	26.7	27.2	27.1	27.3	27.3	27.3	26.7	27.3	26.9	(20)	
	2%	DB	32.8	33.3	33.8	34.0	34.0	33.0	32.6	32.7	32.7	33.0	33.0	33.1	(21)	
		MCWB	26.6	26.4	26.5	27.0	27.2	27.0	26.8	27.0	26.9	26.9	27.0	26.9	(22)	
	5%	DB	32.0	32.5	33.1	33.2	33.0	32.2	31.8	32.0	32.0	32.2	32.3	32.4	(23)	
		MCWB	26.4	26.4	26.5	27.0	27.1	26.8	26.6	26.7	26.6	26.7	26.9	26.7	(24)	
	10%	DB	31.1	31.7	32.3	32.5	32.1	31.4	31.0	31.3	31.1	31.4	31.6	31.7	(25)	
		MCWB	26.2	26.2	26.4	26.9	26.8	26.5	26.3	26.4	26.3	26.5	26.6	26.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	28.0	28.1	28.1	27.7	27.7	27.9	27.8	27.8	27.9	27.8	(27)	
		MCDB	32.7	32.9	33.3	33.6	33.3	32.7	32.7	32.9	32.9	32.8	33.1	33.0	(28)	
	2%	WB	27.2	27.2	27.5	27.7	27.6	27.3	27.1	27.3	27.2	27.2	27.4	27.3	(29)	
		MCDB	32.0	32.3	32.7	33.0	33.0	32.2	32.0	32.1	32.1	32.2	32.4	32.4	(30)	
	5%	WB	26.7	26.8	27.0	27.3	27.2	26.9	26.7	26.9	26.8	26.9	27.0	27.0	(31)	
		MCDB	31.3	31.7	32.2	32.6	32.4	31.7	31.3	31.6	31.4	31.7	31.9	32.0	(32)	
	10%	WB	26.2	26.4	26.6	27.0	26.9	26.5	26.3	26.5	26.4	26.5	26.6	26.6	(33)	
		MCDB	30.7	31.2	31.6	32.1	31.8	31.1	30.7	30.9	30.8	31.1	31.2	31.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	7.8	8.1	7.8	7.3	7.0	7.2	7.1	7.3	7.6	7.6	(35)	
		MCDBR	8.1	8.3	8.6	8.7	8.8	8.1	7.9	8.1	8.2	8.2	8.2	8.4	(36)	
	5% WB	MCWBR	4.1	4.0	3.9	4.2	4.1	4.2	4.1	4.2	4.2	4.1	4.2	4.0	(37)	
		MCWBR	7.6	7.7	8.0	8.3	8.3	7.7	7.6	7.8	7.7	7.8	7.9	8.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.395	0.400	0.411	0.431	0.422	0.406	0.406	0.427	0.431	0.452	0.417	0.404	(40)	
		taud	2.522	2.489	2.464	2.402	2.448	2.500	2.498	2.421	2.398	2.341	2.456	2.492	(41)	
	Edn at Noon		921	924	911	879	869	872	876	871	880	863	892	904	(42)	
		Edh at Noon	107	113	117	122	114	106	107	119	123	129	113	108	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.64	5.28	5.69	5.81	5.29	4.99	4.87	5.01	5.04	5.00	4.91	4.74	(44)	
	RadStd		0.40	0.43	0.40	0.28	0.29	0.22	0.31	0.31	0.39	0.35	0.35	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.24	N/A	N/A	N/A	N/A

Nomenclature: See separate page